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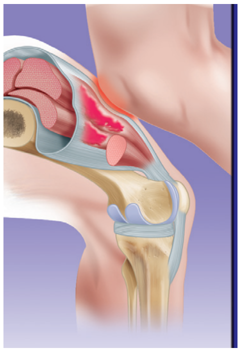

Basics of Taping and Orthotics

Phillip Gribble

TAPING

During physical activity, the integrity of a joint and its associated structures can be compromised, especially during high-risk activities that may introduce injurious mechanisms. Sports medicine clinicians can supplement mechanical support to a joint through a variety of taping techniques using elastic and nonelastic materials. Guidelines have been established that define taping procedures as standard interventions for domains such as "prevention" and "immediate care."[1]

Taping techniques are utilized extensively by athletic trainers and other sports medicine professionals to prevent excessive joint movements, especially in sports known to have a high risk for particular injuries. For example, an individual is at an increased risk for a lateral ankle sprain during participation in the sport of basketball, compared with the sport of baseball.[2,3] Therefore providing additional support to the ankle complex of a basketball player in order to restrict motions outside the anatomical limits is a common preventative measure performed to reduce the risk of an injury to the joint structures. A variety of taping techniques designed for prevention exist that are designed to limit unwanted joint motion or provide biomechanical support to a joint to alleviate discomfort during movement. In addition, a clinician, after evaluating the structures that may have suffered an injurious mechanism during participation in an activity, may determine that the individual can return back to participation with immediate care of supplementing joint stability with a taping technique.

Taping interventions may be employed with either elastic or nonelastic tape using several techniques. Taping materials are available from multiple manufacturers and come in a range of widths and lengths. Elastic tape, which is typically made in widths between 1 and 4 inches (2.5 to 10.2 cm), is used to provide some additional stability while still allowing a large freedom of movement, to secure pads or other protective implements to a body part, or to provide compression. Nonelastic tape can provide more joint support than elastic tape using applications intended to restrict excessive movements, and comes in typical widths between 1 and 2 inches (2.5 to 5.1 cm). Tape may be applied directly to skin that has been shaved or devoid of hair (i.e., sole of the foot), or if the patient does not wish to shave the area that will be taped, a thin layer of foam, called prewrap, may be wrapped around and over the skin surface to be taped.

A newer material used frequently in clinical practice is self-adherent tape and prewrap. This material will adhere to itself, but does not stick to the skin. In addition, it is a very pliable material, with high tensile properties, meaning it can be molded very easily to a body part and still provide a level of mechanical stiffness. The theoretical advantage is that it can provide less skin irritation compared to traditional nonelastic and elastic adhesive tape materials. In addition, the self-adherent prewrap is believed to contribute to the mechanical strength of the taping technique, while regular prewrap does not.

Prophylactic taping techniques can be applied to any part of the body, and there are many variations on the techniques used to apply the nonelastic and elastic materials. Because lower extremity injuries are more prevalent in sports, and these injuries are more likely to be preventable through taping techniques than upper extremity injuries, this overview will focus on some common techniques and associated outcomes related to foot, ankle, and knee taping. It should also be noted that any taping technique may, and often is, modified by a clinician to meet the preventative and treatment needs and comfort of the patient. Therefore the following descriptions are suggested guidelines for some of the more common taping techniques used in sports medicine settings.

FOOT TAPING TECHNIQUES

Plantar Fascia/Arch Taping

Plantar fasciitis or fasciosis is estimated to affect more than 10% of the general population.[4] The most common causes of this pathology relate to a high body mass index, and increased pronation of the midfoot with subsequent flattening of the medial longitudinal arch.[5] Pain persisting along the midfoot extending to the insertion of the plantar fascia at the calcaneus, especially with the first few steps after a period of non-weight-bearing, may benefit from added support to the midfoot. Low-dye and other forms of arch support taping techniques are intended to supplement the structural support of the medial longitudinal arch and the plantar fascia that creates a windlass mechanism needed for normal ambulation.[6]

These taping techniques involve strips of tape that initiate over the insertion of the plantar fascia at the calcaneus or encircle the heel and then extend to the base of each metatarsal head. Elastic or nonelastic tape can be used, with ½ inch widths typically employed. The strips will slightly overlap each other, with

Abstract

Athletic taping techniques are utilized commonly by sports medicine clinicians as an effective intervention to provide joint stability and relieve painful symptoms associated with lower extremity pathologies. While some limitations are associated with these techniques, the evidence supports the use of these materials and techniques, especially in the prevention of injuries. Clinicians need to consider the cost and needs of the patient in selecting the most appropriate techniques and materials to meet the goals of injury prevention and initial treatment these interventions can provide. Orthotics are an effective intervention for treating a variety of lower extremity musculoskeletal overuse injuries by correcting the position of the foot during standing and ambulation. Providing support and/or cushioning to the rearfoot, midfoot, and forefoot may lead to changes in the biomechanical orientation of the entire lower extremity, leading to an alleviation of pain in the feet, knees, and hips. It is important for a trained clinician to conduct a thorough biomechanical evaluation of the feet and the lower extremity kinetic chain to determine the most appropriate form of orthotic to provide comfortable relief.

Keywords

lower extremity
prevention
treatment
ankle
foot
knee

Fig. 33.1 Low-dye taping for foot pain.

Fig. 33.2 Closed basket weave ankle taping technique.

a strip that secures the tape under the metatarsal heads. Versions of low-dye taping typically involve continuing a strip of tape under the medial longitudinal arch, and pulling up toward the lateral surface of the shank, effectively pulling the foot out of a pronated position (Fig. 33.1). Additional strips of elastic tape may be wrapped around the midfoot to support the fan shaped collection of strips supporting the arch of the foot.

In a systematic review, van de Water and Speksnijder[7] examined the short-term effects of taping techniques on treating pain and disability associated with plantar fasciosis. The limited evidence reviewed showed a positive short-term effect (<1 week) on pain reduction with taping techniques compared with control conditions and other interventions, but the evidence was inconclusive on the impact on patient disability. Similar positive short-term effects of taping on stiffness were observed by in combination with iontophoresis, with continued benefits observed after 4 weeks of continued treatment.[8] Relief of pain and symptoms through external support of the foot beyond a week may require additional interventions, such as orthotics, which is discussed later in this chapter.

Toe Taping

Sprains to the first metatarsophalangeal (MTP) joint are common in field sports and can be quite debilitating. "Turf toe," as it is commonly called, can result from a hyperextension or hyperflexion mechanism to the great toe (first MTP joint), which limits dorsiflexion of the great toe and restricts the ability of the patient to push off during ambulation. A taping technique may be employed to resist the painful motion of the MTP joint by looping ½ inch strips of tape to pull the toe into flexion or extension. Additional strips are wrapped around the midfoot to anchor the strips and complete the technique.

Ankle Taping Techniques

The most common anatomical injury among the physically active is to the ankle.[2,3] Subsequently, taping of the ankle for preventive and immediate treatment purposes is the most common technique used in clinical practice. Due to anatomical and biomechanical principles, the lateral ligaments are the most commonly affected structures of the ankle complex. The purpose of a typical ankle taping technique is to limit plantar flexion and inversion of the ankle complex, which are injurious movements associated with a lateral ankle sprain.

The most common technique for providing support to the lateral ankle complex is called the "closed basket weave." Typically performed with 1.5 inch nonelastic tape, the strips of tape are applied as "anchors," "stirrups," and "heel-locks," which will cover the ankle complex from the midshank distally to the tarsals (Fig. 33.2). The combination of strips will position the ankle complex into a dorsiflexed and slightly inverted position. The strips of nonelastic tape may be applied either to skin that has been shaved or over a layer prewrap. In addition, thin square foam pads with petroleum jelly may be placed over the insertion of the Achilles tendon on the calcaneus and over the talus, to reduce friction over these skin areas that may suffer irritation from the crossing of the applied strips of tape.

To provide mild stability and compression to an acutely injured ankle, a clinician may utilize an "open basket weave" taping technique. The sequence of application of the anchor and stirrup strips of nonelastic tape is similar to the closed basket weave technique, but the crisscrossing heel locks are not applied with nonelastic tape, leaving the dorsum of the foot exposed (Fig. 33.3). The heel locks are applied with elastic tape under small amounts of tension, or with an elastic bandage. This technique allows some support to ankle complex with the primary purpose to provide comfortable compression to help minimize the early stages of the inflammatory process.

Knee Taping Techniques

Knee Stability

There are many forms of external prophylactic support for the knee, with rigid braces being the choice most clinicians use for long-term wear and prevention. However, there are taping techniques that can be applied to provide stability to the knee joint during physical activity. Most available knee braces are designed to supplement the cruciate and/or collateral ligaments of the knee. Taping techniques can be applied to mimic or supplement the limitations to knee movement these rigid braces are intended to create.

Fig. 33.3 Open basket weave ankle taping technique.

Fig. 33.4 Knee collateral ligament taping technique.

Fig. 33.5 Knee stability taping technique.

Protection against excessive valgus or varus forces that may threaten the medial and lateral collateral ligaments can be achieved by crisscrossing strips of tape over the medial or lateral surfaces of the knee joint and securing them proximally and distally to the thigh and shank (Fig. 33.4). Additional support can be provided to the cruciate ligaments with this technique by wrapping these crisscrossed strips around the anterior and posterior aspects of the thigh and shank (Fig. 33.5). The angle of each strip is varied in order to comprehensively support the knee joint complex.

Patellofemoral Pain

Patellofemoral pain syndrome (PFPS) is a classification of pathology that presents with anterior knee pain or retropatellar pain. The etiology of PFPS has multiple origins, with many attributed

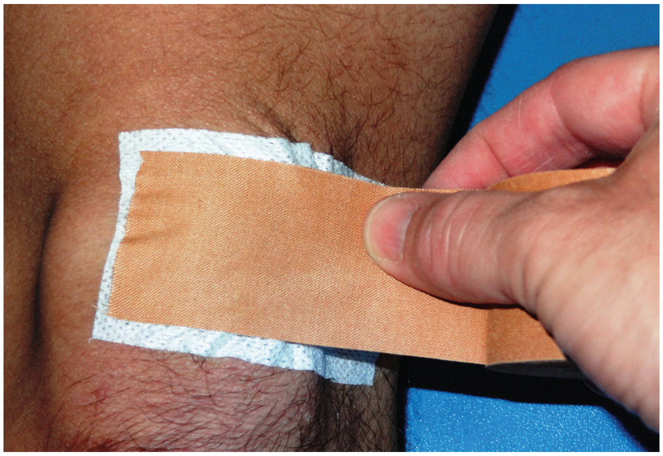

Fig. 33.6 McConnell medial patellar glide taping technique.

to malalignment and poor tracking of the patella in relation to the femoral groove, resulting in pain. One of the many treatments proposed to alleviate the symptoms of PFPS are taping techniques to reposition the patella statically and dynamically. A systematic review by Aminaka and Gribble[9] concluded that patellar corrective taping techniques have positive outcomes for alleviating pain and improving function, but the explanatory mechanisms are not well established by the existing literature. This is likely a result of the complexity and variability of the pathology, which also helps explain why there are so many variations on patellar taping techniques.

A basic tenet of patellar taping techniques is to determine what form of malalignment and maltracking the patient is suffering from (i.e., patella alta, medial rotation, etc.), and then attempt to pull and reposition the patella, using specialized tape, into the opposite direction or combination of directions. The specialized tape is then used to maintain the corrected position, in order to alleviate excessive pressures between the posterior surface of the patella and the trochlear groove of the femur. One popular collection of these techniques was developed by McConnell,[10] but many variations on these techniques exist and may be successful for individual patient needs.[9] In general, these applications are intended to be worn for longer periods of time (days or weeks; see the section below), as the symptoms may persist beyond physical activity into activities of daily living. Subsequently, specialized, and often more expensive, taping materials are utilized. A common technique is to draw the patella medially, using the tape to hold the position statically, and encourage a new movement pattern during dynamic positioning that prevents lateral patellar displacement during a strong contraction of the quadriceps muscle group (Fig. 33.6). Additional adjustments in the tilt, rotation, and vertical placement of the patella can be made with the taping technique.[9,10]

Length of Wear/Length of Effectiveness

The materials used for most athletic taping techniques are not designed for long-term wear. These materials provide a significant amount of stability and movement limitation immediately after application, with most of the published evidence focused on ankle taping techniques.[11–17] However, because they are cloth-based products, exposure to moisture, heat, and tension will quickly reduce the biomechanical stiffness properties of the materials, in as little as 10 minutes after application.[17–20] One study suggests that while both elastic and nonelastic cloth-based tapes will lose tensile strength after 30 minutes of exercise, elastic tape may not lose as much tension as nonelastic tape, and may be more comfortable.[20] Limited investigation on the amount of joint restriction after application and participation in exercise on the newer category of self-adherent tape materials suggests these newer materials may provide more sustainable stability to the ankle, specifically in restricting inversion movement, compared with nonelastic cloth tape.[17] More research is needed to determine how to improve the restrictive properties of taping materials to maximize their effectiveness.

Evidence on the optimal time period of lasting strength of taping materials used in treating foot and knee conditions discussed in this chapter is quite limited. One can only project that the properties of taping materials that have been studied related to ankle taping would apply to the materials if they are used on other joints of the lower extremity and are subjected to similar demands of physical activity. In most cases, taping techniques are designed for short-term wear (a few hours) before being removed. While a taping technique may be applied every day for many weeks or months (i.e., the length of a sport competition season), some techniques, like the Low-Dye and arch support techniques for plantar fasciitis, are typically applied as an intermediary until a more permanent orthotic can be obtained for the patient.

The exception to this are the materials used for patellofemoral corrective taping. These are applied with the intent of the patient wearing the material on the skin potentially for several days in order to receive a benefit of the repositioned patella during all forms of physical and daily living activities. These materials are constructed of more robust material that resists moisture and maintains its mechanical stiffness. The cost of these materials, such as Leukotape (Beiersdorf, Inc., Wilton, CT), is much higher per roll than cloth-based elastic and nonelastic tapes, and would severely limit joint movement, which explains why clinicians may choose not to use these materials in the daily application of most of the taping techniques that have been described in this chapter.

Injury Prevention

There is an ongoing debate in the literature and in clinical practice as to what is the most effective and cost-efficient prophylactic support for reducing injury in the lower extremity. Typically the comparisons are between taping and bracing of the joints of the lower extremity. While taping is a staple among sports medicine clinicians, there is a relatively limited amount of investigation into the efficacy of taping techniques for the prevention of injury, with the majority of this literature focused on ankle injuries.

Ankle Sprains

In general, ankle taping has been shown in clinical studies to reduce the rate of ankle sprain. In an early study, Garrick and Requa[21] reported that taping produced a twofold decrease in ankle sprains in intramural basketball players with a history of an ankle sprain, and a threefold decrease in players who had no history of ankle sprains. The literature continues to suggest that this intervention is effective at reducing ankle injury rates during physical activity and sport participation compared with not wearing a prophylactic support.[22–25] Similarly, a large amount of research supports the use of prophylactic ankle braces for the reduction of ankle sprain incidence.[22–26] However, the existing evidence does not seem to support a difference in the effectiveness of ankle taping versus ankle bracing, meaning both forms of prevention are useful in reducing ankle sprain incidence.[22–25,27] Short- and long-term cost of materials may be an important deciding factor, which will be discussed later in the chapter.

Plantar Fasciitis/Fasciosis

While low-dye and other taping techniques are used successfully in clinical practice to alleviate excessive pronation and plantar fasciitis and fasciosis symptoms, the published evidence is quite limited. Typically this treatment is used initially to alleviate symptoms, but long-term use commonly is avoided in favor of other forms of mechanical correction and support to the foot through orthotics and other forms of tissue treatment to reduce inflammation and irritation. Supporting the clinical use of this technique, a systematic review concluded that taping techniques are a short-term effective treatment for pain associated with plantar fasciitis/fasciosis, but the evidence is inconclusive for supporting these techniques to improve disability.[7] Similarly, a more recent investigation stated that low-dye taping was able to decrease patient-reported pain during walking and jogging immediately after application, but long-term effects were not assessed.[28] Therefore it appears that these taping techniques can provide an immediate and short-term resolution of pain in patients with plantar fasciitis.

Patellofemoral Pain

Taping techniques for PFPS continue to be a popular intervention in clinical practice. The repositioning of the patella may be able to change the movement patterns of the lower extremity, but the primary outcome variable that has been studied is alleviation of pain. Previous systematic reviews of the literature support the use of patellofemoral repositioning taping techniques to alleviate pain in short- and long-term periods,[9,29] while another review suggests the literature is too inconclusive to make a clinical recommendation.[30] While continued work is needed to verify how this category of taping technique is able to alleviate symptoms of patellofemoral pain, it appears it may represent a viable intervention for patients suffering this pathology.

Criticisms of Taping Techniques

One criticism of taping techniques that is often raised is that the applications will limit performance during physical activity. Cordova et al.[31] concluded in a meta-analysis that external ankle supports, including taping, do not significantly hinder most measures of physical performance. Most patients will be cognizant of the taping application immediately upon application, due to the intended restrictive or supportive nature of the technique. This may be a positive effect, as these techniques are likely providing stimulation to cutaneous receptors, which may be supplementing the neuromuscular control systems that will provide dynamic stability to the joint.[32,33]

Length of Effectiveness

As mentioned previously in this chapter, the materials used in athletic taping techniques do not maintain structural stiffness for very long after application.[11,12,16,18,19] While this is a legitimate criticism of these materials, the evidence shows that the use of taping techniques can reduce injury rates[22–25,27] and reduce pathology associated pain.[7–9,28] Therefore in spite of a preclusion to lose mechanical stiffness, taping techniques are an effective intervention. A likely explanation is the increased afferent signaling that taping applied to the skin generates.[32,33] An increase in proprioceptive information may be generating feedforward responses in the central nervous system that contribute to improved muscle reaction and dynamic stability, and subsequently a decreased in injurious positions and alleviation of pain.

Cost Effectiveness

Most of the existing literature supports the effectiveness of all forms of ankle prophylactic support for the prevention of ankle sprains. However, the cost of materials and the time to apply them must be considered by clinicians. Olmsted et al.[24] showed that ankle taping and bracing were effective at reducing ankle sprains, especially in athletes with a history of ankle instability. However, during a full athletic competition season, taping an ankle is three times more expensive than bracing an ankle.[24] Similarly, Mickel et al.[27] found ankle taping and bracing to be equally effective at reducing ankle sprains in high school football players, but taping an ankle every day for an entire season would cost more than providing a brace that could be applied every day. In addition, taping requires substantially more time by the clinician to administer the taping techniques every day.[24]

Taping Summary

Athletic taping techniques are utilized commonly by sports medicine clinicians as an effective intervention to provide joint stability and relieve painful symptoms associated with lower extremity pathologies. While some limitations are associated with these techniques, the evidence supports the use of these materials and techniques, especially in prevention of injuries. Clinicians need to consider the cost and needs of the patient in selecting the most appropriate techniques and materials to meet the goals of injury prevention and initial treatment these interventions can provide.

ORTHOTICS

Participation in physical activity, such as running, increases the chance of lower extremity overuse injuries throughout the kinetic chain of the ankle, knee, hip, and low back. Many of these ailments can be attributed to malalignments and improper biomechanics of the foot. Intrinsic, structural abnormalities in the medial longitudinal arch (pes planus, pes cavus), as well as leg length discrepancies, may impact the static alignment of the lower extremity and low back, which can translate into inefficient force transmission during physical activity. Examining the alignment of the forefoot, midfoot, and rearfoot with the entire lower extremity kinetic chain may help elucidate the onset of static and dynamic abnormalities throughout the lower extremity. Foot structural malalignments may be creating problems elsewhere in the kinetic chain, or may be an adaptation to a biomechanical or neuromuscular problem proximal to the foot. A common and effective intervention is to correct malalignments of the foot with the use of an orthotic shoe insert to optimize force transmission from the foot into the leg and back during contact with the ground.[34–36] It is imperative that the clinician considers individual patient needs to ensure the selection of the most appropriate orthotic.

Evaluating the Foot

The primary goal of an orthotic is to create an optimal and comfortable orientation between the three areas of the foot (forefoot, midfoot, and rearfoot) in order to efficiently attenuate forces experienced during contact with the ground. Much of the attention of creating an optimal foot position centers on creating a subtalar neutral position, which means the talus is positioned in a neutral position within the ankle mortise and in relation to the calcaneus, while optimizing the integrity of medial longitudinal arch.

Examining the foot structure should involve observation of the orientation of forefoot, midfoot, and rearfoot in relation to each other in weight-bearing and non-weight-bearing, and how any changes in these foot positions affect the orientations of the lower leg, thigh, and pelvis. Root et al.[37] originally described a procedure to prescribe orthotics based on examining the static orientation of the rearfoot and finding a position of subtalar neutral. Later, Landorf et al. showed that prescription and construction of orthotics varies significantly among clinicians.[38] More recent approaches to prescribing orthotics suggest a much more comprehensive evaluation of the entire lower limb during static and dynamic assessments to determine the needs of the patient and how an orthotic may be able to help correct alignment issues and reduce pain.[36,39,40] Once the evaluation is complete, the clinician can decide what factors will be needed in the orthotic (i.e., motion control, cushioning, medial vs. lateral support, etc.) and which category of orthotic will be best for the patient's needs.

While multiple diagnoses may emerge from an evaluation of the possible foot and lower limb orientations, two of the more common categories of general foot malalignments, or foot types, are pes planus, or excessive pronation/flat arch, and pes cavus, or excessive supination/high arch. Pes planus typically presents with rearfoot and forefoot varus, causing the medial longitudinal arch to flatten excessively to the ground during weight acceptance. This foot type may benefit from more support on the medial side of the foot. Conversely, a pes cavus foot commonly will have contributions of rearfoot and forefoot valgus, limiting the ability of the medial longitudinal arch to flatten normally during gait. This condition may benefit from more orthotic support under the lateral surface of the foot. While this describes these conditions and approaches to orthotic intervention in general, it is important for a trained clinician to examine the foot thoroughly to determine the individual needs of the patient.

Fig. 33.7 Soft, semi-rigid, and rigid full-length orthotics *(bottom view)*.

Types of Orthotics and Materials

Foot orthotics can be constructed from a variety of soft, semi-rigid, and rigid materials, and may be categorized as molded, or custom-made, or nonmolded, or over-the-counter (Fig. 33.7). As emphasized previously, the selection of materials is specific to the needs of the patient, as determined by the evaluation of the clinician. In general, softer materials provide more cushioning, whereas rigid materials provide more stability and control of motion. Semirigid orthotics are intended to provide a combination of cushioning and motion control. Rigid orthotics are moldable and are custom-made to the meet the specifications of the patient's foot, whereas the soft orthotics are not designed to be altered, and can be used "off the shelf" immediately by the patient without having to wait for custom modifications. Semirigid orthotics provide an option that is off the shelf, and can be utilized as is, but also may be altered through minor customization by a clinician. Soft orthotics have a much shorter lifespan than rigid and semirigid orthotics because the inherent thickness and stiffness of the material in soft orthotics is much less. Subsequently, rigid and semirigid orthotics have higher associated costs because of more robust material and the added expense of professional construction and/or modification of the product.

If a decision is made by the evaluating clinician that the patient needs corrective orthotics to provide foot stability and/or motion control, either a semirigid or rigid orthotic will be selected and constructed. Treating basic foot abnormalities such as an excessive pes planus or pes cavus foot may be successful with a semirigid orthotic. These are constructed of material that provides shock absorption and motion control. Semirigid orthotics are manufactured in standard shoe sizes and may be worn as is by the patient, but are designed to allow a clinician to attach additional support, or "posting," with foam or thermoplastic materials to create more medial or lateral stability as needed. Semirigid orthotics offer a good solution for a customizable motion control orthotic, but depending on the demands and the degree of foot malalignment, the patient may benefit from a rigid, custom-made orthotic.

Rigid orthotics are designed to provide the greatest amount of motion control and stability. Constructed of acrylic plastics and other composite materials, these orthotics require a lengthier process to produce. First, the clinician will make a cast impression of the patient's foot after positioning the patient in subtalar neutral and midtarsal joint stabilized against the rearfoot (Fig. 33.8). As stated earlier, the methods for determining the optimal alignment of the foot differ among clinicians,[36–40] and a comprehensive discussion of the methods utilized in clinical practice is outside the scope of this chapter. Nonetheless, once a cast of the foot is made, the cast is used to make a positive mold of an orthotic with the rigid materials. While the construction of the orthotic may be performed in a clinic, often the clinic will utilize a service that receives the cast mold, manufactures the orthotic, and sends the orthotic back to the clinic for distribution to the patient (Fig. 33.9).

Fig. 33.8 Orthotic mold cast for creation of rigid orthotic.

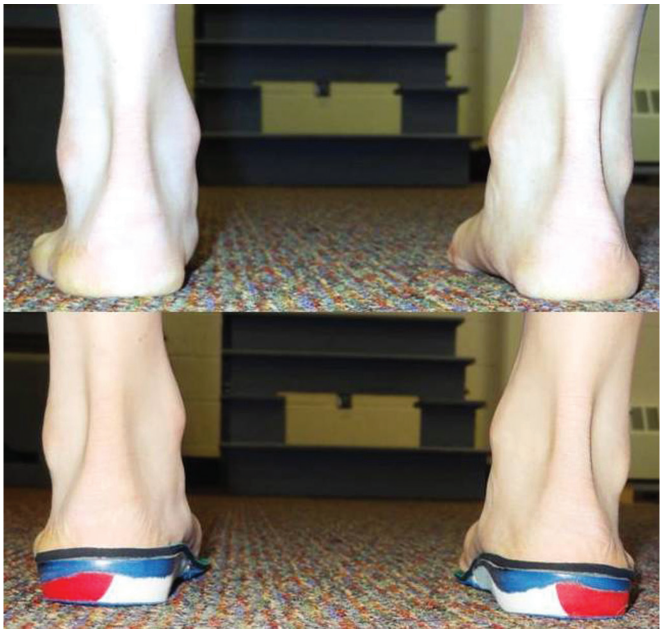

Fig. 33.9 Correction of pes planus with custom-made rigid orthotic.

Outcomes for Common Overuse Issues

Orthotics continue to be prescribed in the treatment of lower extremity musculoskeletal pathologies because they have a good rate of successful outcome. In controlled laboratory studies, orthotics are able to improve balance in subjects with cavus feet[41] and after suffering a lateral ankle sprain.[42,43] In addition, it is suggested orthotics are able to treat excessive pronation by decreasing eversion excursion and velocity during walking[44] and running.[45]

Running is an activity associated with a high rate of overuse injuries, and orthotics have been associated with successful reductions in a variety of injuries, including anterior knee pain, plantar fasciitis, and Achilles tendonitis.[46,47] One review of literature concluded that with an orthotic, 64% to 95% of runners will experience moderate to full alleviation of pain caused by running.[48] More recently, a systematic review reported that custom-made orthotics were very effective at reducing pain associated with pes cavus, and moderately effective at reducing pain associated with hallux valgus and in juvenile idiopathic arthritis.[49] Finally, a meta-analysis concluded that orthotics are successful at reducing pain and improving function in patients with plantar fasciitis.[50]

Unlike the taping techniques described earlier in this chapter, orthotics are typically used as a treatment, rather than as a preventative measure. A recent randomized-controlled trial showed that among military cadets in Finland, orthotics did not prevent overuse injuries during 8 weeks of military training compared with a control condition.[51] In a contrasting randomized-controlled trial, military cadets in the United Kingdom who were randomly given orthotics experienced fewer lower extremity overuse injuries during 7 weeks of military training compared with cadets who did not receive preventative orthotics.[52] With the limited evidence that exists, there seems to be potential for orthotics to prevent lower extremity injuries, but it is difficult to make definitive conclusions, especially for application among populations of varying levels of physical activity.

While most of the research suggests that orthotics are an effective intervention for correcting skeletal malalignments, it is also suggested that orthotics may enhance the role of cutaneous receptors,[35,53] leading to improvements in balance and function. In addition, clinicians should not ignore the importance of patient comfort when utilizing these interventions.[35] Vicenzino et al.[36,40] advocate that simply correcting the foot position does not necessarily alleviate the biomechanical problem in the lower limb; and an orthotic based only on the foot position may not result in the most comfortable orthotic for the patient. Therefore outcomes with orthotic intervention are likely to be improved by considering a comprehensive lower extremity evaluation and the comfort of the patient.

SUMMARY

Orthotics are an effective intervention for treating a variety of lower extremity musculoskeletal overuse injuries by correcting the position of the foot during standing and ambulation. Providing support and/or cushioning to the rearfoot, midfoot, and forefoot may lead to changes in the biomechanical orientation of the entire lower extremity, leading to an alleviation of pain in the feet, knees, and hips. It is important for a trained clinician to conduct a thorough biomechanical evaluation of the feet and the lower extremity kinetic chain to determine the most appropriate form of orthotic to provide comfortable relief.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Aminaka N, Gribble P. A systematic review of the effects of therapeutic taping on patellofemoral pain syndrome. *J Athl Train*. 2005;40(4):341–351.

Level of Evidence:
Ia

Summary:
This systematic review investigated the efficacy of patellar taping on pain control, patellar alignment, and neuromuscular control (i.e., vastus medialis oblique activation, knee extensor moment, etc.) in subjects with patellofemoral pain syndrome. Although patellar taping seems to reduce pain and improve function in people with patellofemoral pain syndrome during activities of daily living and rehabilitation exercise, strong evidence to identify the underlying mechanisms is still not available.

Citation:
Dizon J, Reyes J. A systematic review on the effectiveness of external supports in the prevention of inversion ankle sprains among elite and recreational players. *J Sci Med Sport*. 2010;13:309–317.

Level of Evidence:
Ia

Summary:
This study was conducted to evaluate the effectiveness of external ankle supports in the prevention of inversion ankle sprains and identify which type of ankle support was superior to the other. The main significant finding was the reduction of ankle sprain by 69% with the use of ankle brace and reduction of ankle sprain by 71% with the use of ankle tape among previously injured athletes.

Citation:
Vicenzino B, Collins N, Cleland J, et al. A clinical prediction rule for identifying patients with patellofemoral pain who are likely to benefit from foot orthoses: a preliminary determination. *Br J Sports Med*. 2010;44:862–866.

Level of Evidence:
Ib

Summary:
This post-hoc analysis of a RCT intended to develop a clinical prediction rule to identify patients with patellofemoral pain (PFP) who are more likely to benefit from foot orthoses. Age, height, pain severity, and midfoot morphometry were identified as possible predictors of successful treatment of PEP with foot orthoses, thereby providing practitioners with information for utilizing foot orthoses in patients with PFP.

Citation:
Hawke F, Burns J, Radford J, et al. Custom-made orthoses for the treatment of foot pain. *Cochrane Database Syst Rev*. 2008;(3):CD006801.

Level of Evidence:
Ia

Summary:
This systematic review from the Cochrane library evaluated the effectiveness of custom foot orthoses for different types of foot pain. There was strong evidence in support of using orthoses in

the treatment of painful pes cavus feet, and moderate evidence in treating juvenile idiopathic arthritis, rheumatoid arthritis, plantar fasciitis, and hallux valgus.

Citation:

Franklyn-Miller A, Wilson C, Bilzon J, et al. Foot orthoses in the prevention of injury in initial military training: a randomized controlled trial. *Am J Sports Med*. 2011;39(1):30–37.

Level of Evidence:

Ib

Summary:

Custom-made foot orthoses were found to be an effective preventative intervention to reduce overuse lower limb injury rates in an "at risk'" military trainee population.

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